A CASE-STUDY BASED REPORT

ON

A Machine Learning Based Case Study on Water Quality Classification Using Random Forest and Support Vector Machine

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

S. Ketana Sindhu 2520030487

B. Chandrika 2520030041

Under the Esteemed guidance of

Mr. Shaik Asif

Asst.Professor, Dept. of CSE



(DEEMED TO BE UNIVERSITY)

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

INDEX

Abstract.....**1**

Introduction.....**2 - 6**

Background.....
Motivation for the Project.....
Overview of the Work Carried Out.....
Scope and Responsible-Use Statement.....

Problem Statement.....**7-9**

Case-Study Scenario.....
Central Case-Study Question.....
Formal Problem Definition.....
Objectives that Guide the Study.....
Assumptions and Limitations.....

Dataset and Exploratory Data Analysis.....**10-13**

About the Dataset.....
Purpose of EDA.....
Preparation Steps Actually Performed.....
EDA Exploration - Implementation.....
Summary of EDA Findings.....

Algorithm Selection and Implementation.....**14-16**

Conclusion.....**20-21**

Signature of the Team members

ABSTRACT

Water quality is an important factor affecting public health and environmental sustainability. This project presents a machine-learning-based water quality classification system that predicts whether a water sample is potable or non-potable using its physical and chemical characteristics. The project uses the Water Potability dataset containing 3,276 samples and nine input features: pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, and Turbidity.

Two supervised machine learning algorithms, Random Forest and Support Vector Machine (SVM), are implemented and compared. The workflow includes dataset inspection, exploratory data analysis, handling missing numeric values through median imputation, feature scaling where required, model training, and performance evaluation. The dataset is divided into 80% training data and 20% testing data using a stratified split. Model performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and five-fold cross-validation.

On the test dataset, Random Forest achieved an accuracy of 67.23%, while SVM achieved 66.92%. Random Forest obtained a higher F1-score, whereas SVM achieved higher precision. The project also provides a web-based interface through which users can enter water-quality measurements, view predictions, explore model performance, and examine dataset information. The system is intended for preliminary educational analysis and does not replace laboratory testing or official water-safety assessments.

1.INTRODUCTION

1.1 Background

Water quality classification is an important application of data analysis and machine learning. Water contains various physical and chemical characteristics that influence its suitability for drinking. Parameters such as pH, hardness, dissolved solids, chloramines, sulphate, conductivity, organic carbon, trihalomethanes, and turbidity provide information about the characteristics of a water sample.

Traditional water-quality assessment may involve laboratory procedures, specialised equipment, and expert interpretation. Machine learning provides a way to analyse several measurements together and identify patterns from previously labelled data.

In this project, a supervised machine learning approach is used to classify water samples into two categories: potable and non-potable. Random Forest and Support Vector Machine algorithms are trained and evaluated to compare their classification performance. A web interface is also developed to make the system easier to demonstrate and use.

1.2 Motivation for the Project

The motivation behind this project is to explore how machine learning can support preliminary water-quality classification using multiple input parameters.

The major motivations are:

- To understand how physical and chemical water measurements can be used as machine learning features.
- To apply supervised learning to a real-world classification problem.
- To compare Random Forest and Support Vector Machine using consistent evaluation metrics.
- To understand the importance of data preprocessing, exploratory data analysis, and model evaluation.
- To provide an accessible web interface for entering sample measurements and viewing model predictions.

- To identify the limitations of machine learning predictions in an application related to public health.

The project also provides practical experience in developing a complete machine learning workflow, from dataset preparation to model evaluation and application integration.

1.3 Overview of the Work Carried Out

The project was carried out through the following stages:

1. Dataset collection: The Water Potability dataset was selected for the classification task.
2. Data inspection: The dataset dimensions, feature names, data types, missing values, and target distribution were examined.
3. Exploratory Data Analysis: Histograms, a missing-value heatmap, a correlation heatmap, and boxplots were used to explore the dataset.
4. Data preprocessing: Missing numeric values were handled using median imputation, and feature scaling was applied where required.
5. Model development: Random Forest and Support Vector Machine classifiers were trained using the prepared data.
6. Model evaluation: Accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and five-fold cross-validation were used to evaluate the models.
7. Results analysis: The model results were compared, and Random Forest feature importance was examined.
8. Web application development: A web interface was created with Prediction, Analytics, Models, and Dataset sections.

1.4 Scope and Responsible-Use Statement

The scope of this project is to develop and evaluate a machine learning system that classifies water samples based on nine available features. The system demonstrates how data preprocessing, classification algorithms, and evaluation metrics can be integrated into an application.

The system provides preliminary predictions based on patterns learned from the dataset. It does not analyse every possible contaminant and cannot guarantee that water is safe for drinking. The predictions must not be used as a substitute for laboratory testing, professional assessment, or applicable drinking-water standards.

2. Problem Statement

2.1 Case – Study Scenario

Consider a situation in which a water sample has been measured for parameters such as pH, hardness, dissolved solids, chloramines, sulphate, conductivity, organic carbon, trihalomethanes, and turbidity.

Interpreting these measurements together can be challenging without a systematic analytical approach. A machine learning model can learn patterns from labelled water samples and use those patterns to predict whether a new sample belongs to the potable or non-potable category.

The proposed Water Quality Classification System uses Random Forest and Support Vector Machine to generate predictions from the input measurements. The results are presented through a web interface so that users can explore predictions and model-performance information.

2.2 Central Case – Study Question

How effectively can Random Forest and Support Vector Machine classify water samples as potable or non-potable using nine water-quality features, and how do their performance metrics compare?

2.3 Formal Problem Definition

The problem is formulated as a supervised binary classification task.

Each water sample is represented by nine input features:

- pH
- Hardness
- Solids
- Chloramines
- Sulphate
- Conductivity
- Organic Carbon

- Trihalomethanes
- Turbidity

The target variable is named Potability. A value of 1 represents potable water, while a value of 0 represents non-potable water.

Let the input feature vector be represented by X and the target class by Y. The objective is to learn a classification function from labelled training samples that predicts the target class for unseen samples.

Two classification algorithms, Random Forest and Support Vector Machine, are implemented. Their performance is compared using appropriate classification metrics.

2.4 Objectives that Guide the Study

The primary objectives of the project are:

1. To analyse a water-quality dataset containing 3,276 samples and nine input features.
2. To identify missing values, examine feature distributions, and study relationships among the input variables.
3. To prepare the dataset for machine learning using appropriate preprocessing techniques.
4. To train Random Forest and Support Vector Machine classifiers.
5. To compare the models using accuracy, precision, recall, F1-score, and ROC-AUC.
6. To use five-fold cross-validation to examine model performance across different data folds.
7. To identify influential features using Random Forest feature importance.
8. To develop a web interface for predictions, analytics, model information, and dataset exploration.
9. To discuss the limitations and responsible use of the developed classification system.

2.5 Assumptions and Limitations

The project uses the following assumptions and acknowledges these limitations:

- The dataset's existing Potability labels are treated as the reference labels for training and evaluation.
- Only the nine available features are used for classification.
- The dataset contains more non-potable samples than potable samples.
- The model learns patterns from the available dataset and may not generalise equally well to every geographical location or water source.
- Some numeric features contain missing values, requiring preprocessing.
- Model performance depends on the quality and representativeness of the dataset.
- The reported evaluation results do not guarantee the safety of a real-world water sample.
- The models have limited recall for the potable class, so some genuinely potable samples may be classified incorrectly.
- Laboratory testing and appropriate regulatory assessment remain necessary for actual drinking-water safety decisions.

3. DATASET AND EXPLORATORY DATA ANALYSIS

3.1 About the Dataset

The project uses the Water Potability dataset commonly distributed through Kaggle. The dataset contains information about water samples and a binary target indicating potability.

The dataset consists of 3,276 records and 10 columns, including nine input features and one target column named Potability.

The input features describe different physical and chemical properties of the water samples. The target variable indicates whether each sample is labelled potable or non-potable.

Dataset summary:

- Total samples: 3,276
- Input features: 9
- Target variable: Potability
- Total columns: 10
- Non-potable samples: 1,998
- Potable samples: 1,278
- Training samples: 2,620
- Testing samples: 656
- Train-test split: 80:20, stratified

The class distribution is approximately 60.99% non-potable and 39.01% potable.

Description of input features

Feature	Description
pH	Measures the acidity or alkalinity of water.
Hardness	Represents water hardness.
Solids	Represents total dissolved solids.
Chloramines	Represents chloramine concentration.

Sulfate	Represents sulfate concentration.
Conductivity	Measures electrical conductivity.
Organic Carbon	Represents organic carbon measurement.
Trihalomethanes	Represents trihalomethane concentration.
Turbidity	Represents the cloudiness of water.

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) is performed to understand the structure, quality, distribution, and relationships of the dataset before model training.

The main purposes of EDA in this project are:

- To understand the number of samples, input features, and target classes.
- To identify missing values in the dataset.
- To examine the distributions and ranges of the numeric features.
- To explore relationships between numeric features using correlation analysis.
- To inspect the spread of selected features and identify potential outliers.
- To understand the target-class imbalance.
- To guide the selection of suitable preprocessing steps.

EDA helps identify potential data-quality issues before the classification models are trained.

3.3 Preparation Steps Actually Performed

The following preparation steps were included in the project workflow:

1. Dataset inspection

The dataset was loaded and examined to understand its dimensions, column names, data types, and descriptive statistics.

2. Missing-value analysis

Missing values were checked to identify incomplete observations. A missing-value heatmap was used to visualise the missingness pattern.

3. Feature-distribution analysis

Histograms were used to examine the distributions of numeric features, including Hardness, Solids, Chloramines, Sulphate, Conductivity, and Organic Carbon.

4. Correlation analysis

A correlation heatmap was used to explore relationships among numeric features.

5. Boxplot analysis

Boxplots were generated for selected features, including Hardness, Solids, Chloramines, and Sulfate, to examine variability and potential outliers.

6. Missing-value handling

Median imputation was used for missing numeric values in the model-preparation workflow.

7. Feature scaling

Feature scaling was applied where required, particularly for the Support Vector Machine model, because its performance can be affected by differences in feature scales.

8. Feature-target separation

The nine input features were separated from the target variable, Potability.

9. Train-test split

The dataset was divided into 80% training data and 20% testing data using a stratified split, resulting in 2,620 training samples and 656 testing samples.

3.4 EDA Exploration – Implementation

The exploratory analysis was implemented using Python data-analysis and visualisation tools.

The missing-value heatmap was used to inspect incomplete observations. Histograms were used to understand the distributions of numeric measurements. The correlation heatmap was used to study relationships between the numeric

features. Boxplots were used to examine the spread, median, variability, and potential outliers in selected measurements.

The EDA findings helped guide the preprocessing and modelling stages. In particular, missing numeric values required imputation, and the different feature ranges supported the use of scaling for SVM.

The presence of a potential outlier does not necessarily mean that a measurement is incorrect. Therefore, unusual observations should not automatically be removed without sufficient justification.

3.5 Summary of EDA Findings

The main findings from the exploratory analysis are:

1. The dataset contains 3,276 samples and nine input features.
2. The target variable has two classes: potable and non-potable.
3. There are 1,998 non-potable samples and 1,278 potable samples.
4. The target distribution is imbalanced, with non-potable samples forming approximately 60.99% of the dataset.
5. Missing numeric values require an appropriate imputation step before model training.
6. The numeric features have different distributions and measurement ranges.
7. The boxplots indicate that some observations lie far from the central distribution and may warrant further investigation.
8. The EDA findings support careful preprocessing and the use of multiple evaluation metrics instead of accuracy alone.

4. ALGORITHM SELECTION AND IMPLEMENTATION

4.1 Random Forest Algorithm

Random Forest is a supervised ensemble learning algorithm that combines the predictions of multiple decision trees. Each tree contributes to the final classification, allowing the model to capture non-linear relationships and interactions among the input features.

Random Forest was selected because it can work with multiple numeric features and provides feature-importance values that help interpret the model.

The final Random Forest parameters used in the project are:

Parameter	Value
Number of estimators	400
Maximum depth	20
Minimum samples per leaf	2
Minimum samples to split	5
Class weight	Balanced

The balanced class-weight setting assigns weights to classes to account for the unequal class distribution during model training.

4.2 Support Vector Machine Algorithm

Support Vector Machine is a supervised learning algorithm that identifies a decision boundary separating different classes. The project uses an RBF (Radial Basis Function) kernel, which allows the model to learn a non-linear decision boundary.

Feature scaling is important for SVM because the input features have different numerical ranges.

The final SVM parameters used in the project are:

Parameter	Value

C	1
Kernel	RBF
Gamma	0.1
Class weight	None

Both Random Forest and SVM were trained and evaluated using the prepared dataset.

4.3 Training and Evaluation Procedure

The model development process followed these steps:

1. Separate the input features and target variable.
2. Prepare missing numeric values using median imputation.
3. Apply feature scaling where required.
4. Divide the data into training and testing sets using an 80:20 stratified split.
5. Train the Random Forest and SVM classifiers using the training data.
6. Generate predictions on the held-out test data.
7. Calculate accuracy, precision, recall, F1-score, and ROC-AUC.
8. Examine confusion matrices and ROC curves.
9. Perform five-fold cross-validation to examine performance across different folds.
10. Compare the models and present the results through the web interface.

4.4 Evaluation Metrics

Accuracy: Measures the proportion of all predictions that are correct.

Precision: Measures the proportion of predicted positive samples that are actually positive. In this project, positive class 1 represents potable water.

Recall: Measures the proportion of actually potable samples correctly identified by the model.

F1-score: Combines precision and recall using their harmonic mean.

ROC-AUC: Measures how effectively a model ranks positive samples above negative samples across different classification thresholds.

Confusion matrix: Summarises correct and incorrect predictions for each class.

These metrics provide a broader view of model performance. In particular, recall is important to examine because failing to identify a potable sample affects the interpretation of the model's predictions. Nevertheless, none of these metrics alone establishes whether real-world water is safe to drink.

5. RESULTS AND DISCUSSION

5.1 Final Test Results

The following results were obtained on the held-out test dataset.

Evaluation Metric	Random Forest	SVM
Accuracy	67.23%	66.92%
Precision	65.19%	70.10%
Recall	34.38%	26.56%
F1-score	45.01%	38.53%
ROC-AUC	66.20%	65.05%

Random Forest achieved an accuracy of 67.23%, slightly higher than the SVM accuracy of 66.92%. Random Forest also achieved higher recall and F1-score on the test dataset.

SVM achieved higher precision, at 70.10%, compared with 65.19% for Random Forest. However, its recall was lower.

The relatively low recall values indicate that both models fail to identify a substantial proportion of the samples belonging to the potable class at the selected classification threshold. Therefore, the results show that the models have limitations and require further improvement before any real-world application could be considered.

5.2 Five – Fold Cross – Validation Results

Five-fold cross-validation was used to evaluate how model performance varied across different folds of the data.

Metric	Random Forest	SVM
Accuracy	68.13% ± 1.60%	67.90% ± 1.47%
Precision	65.80% ± 2.91%	72.25% ± 4.17%
Recall	37.96% ± 3.84%	28.77% ± 3.33%

F1-score	48.08% ± 3.62%	41.06% ± 3.80%
ROC-AUC	69.32% ± 1.21%	70.13% ± 1.72%

Random Forest achieved slightly higher mean accuracy, recall, and F1-score in the recorded cross-validation results. SVM achieved higher mean precision and ROC-AUC.

These results demonstrate that the choice of model depends on the metric being considered. No single model should be declared universally superior based only on one evaluation measure.

5.3 Random Forest Feature Importance

Random Forest provides feature-importance values that indicate the relative contribution of input features to its tree-based predictions.

The final feature-importance values obtained in the project are:

Feature	Importance
pH	0.137464
Sulfate	0.132642
Hardness	0.122545
Chloramines	0.116695
Solids	0.111971
Conductivity	0.098762
Turbidity	0.095348
Organic Carbon	0.093043
Trihalomethanes	0.091531

According to the fitted Random Forest model, pH has the highest feature-importance value, followed by Sulfate and Hardness. Trihalomethanes has the lowest importance among the nine features in this model.

These values describe the model's learned feature usage. They do not establish that one feature causes water to be potable or non-potable.

5.4 Web Application Description

The project includes a web application named WaterIQ for demonstrating the water-quality classification system.

The application contains the following sections:

Prediction: Allows users to enter the nine water-quality measurements and request predictions from the classification system.

Analytics: Presents model evaluation metrics and visual comparisons of the models.

Models: Displays information about Random Forest and SVM, their recorded performance metrics, and their final parameter configurations.

Dataset: Presents the dataset summary, sample counts, target-class distribution, and descriptions of the input features.

The application uses a frontend interface and a Python backend API to connect user inputs with the trained machine learning models. The project also stores trained model files and result artifacts for use by the application.

6. CONCLUSION

This project developed a machine-learning-based water quality classification system using the Water Potability dataset, Random Forest, and Support Vector Machine. The dataset contains 3,276 samples and nine input features. The project covered dataset inspection, exploratory data analysis, missing-value handling, feature scaling where required, model training, evaluation, cross-validation, and web application development.

The final test results showed that Random Forest achieved 67.23% accuracy, while SVM achieved 66.92%. Random Forest obtained higher recall and F1-score on the held-out test set, whereas SVM achieved higher precision. The cross-validation results also showed differences in performance depending on the metric considered.

Although both algorithms demonstrated the ability to classify water samples using the available measurements, their recall values were relatively low. This indicates that the current models have limitations and may fail to correctly identify many samples belonging to the potable class.

The project successfully demonstrates the application of supervised machine learning to a water-quality classification problem and provides a web interface for exploring predictions and evaluation results. However, the system should be considered a preliminary educational tool rather than a definitive water-safety assessment. Laboratory testing and appropriate regulatory procedures remain necessary for determining whether water is safe for drinking.

6.1 FUTURE SCOPE

The project can be improved in the following ways:

1. Evaluate additional machine learning algorithms and systematic hyperparameter optimisation techniques.
2. Investigate class-imbalance handling and classification-threshold selection to improve the balance between precision and recall.
3. Test the models on independent datasets collected from different geographical locations.

4. Improve the explanation of individual predictions and communicate uncertainty more clearly.
5. Improve the user interface and add further model-comparison visualisations.
6. Integrate verified sensor measurements only after suitable calibration, validation, and reliability testing.
7. Evaluate the system against relevant water-quality standards before considering any operational use.